Elements and Compounds

Elements are said to be _____ substances and are made up of only one kind of atom. Elements cannot be broken down into simpler substances.

There are ___ naturally occurring elements and many man-made elements. Each element has its own unique _____.

Elements can combine together chemically to form _____.



H_2O
(water)

Compound Formulae

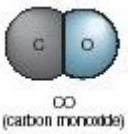
Elements combine in predictable ways to make compounds. For example, a **water** molecule is always formed when **one oxygen** atom combines with **two hydrogen** atoms. The chemical formula for water is _____.

For each of the compounds below, state what elements the compound contains and how many of each.

Formula of Compound	**Types of Elements it Contains**	**Number of Atoms of Each Element**
Carbon Dioxide CO_2		
Methane CH_4		
Calcium Chloride $CaCl_2$		
Copper Sulphate $CuSO_4$		

Naming Compounds

There are some rules for naming chemical compounds.



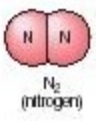
CO
(carbon monoxide)

If the compound contains a metal atom and a non-metal atom,

- Write the name of the **metal** first
- Write the name of the **non-metal** second and change the ending to **–ide**

For example, table salt contains one sodium atom and one chlorine atom, NaCl. The name of this compound is _____ _____.

Try writing the names for these compounds.



N_2
(nitrogen)

KCl _____ Fe_2O_3 _____

NaI _____ Na_2O _____

CaO _____ AgI _____

MgS _____ $NiCl_2$ _____

If the compound contains only non-metal atoms, you need to use **prefixes** to describe how many atoms of each element are in the compound.



mono – one **di** – two **tri** – three **tetra** – four

- The elements are **named in the order they appear** in the formula.
- The **prefix** is put in front of each element name to show **how many** are in the compound.
- The **ending** of the last element is changed to **–ide**.
- If there is only **one** atom of the first element, **no prefix** is needed.

For example, the formula for water is H_2O. It contains two hydrogen atoms and one oxygen atom. Its scientific name is _____ _____.



Try writing the names for these compounds.

CO_2 _____ NF_3 _____

SO_3 _____ H_2O_2 _____

NH_3 _____ H_2S _____

N_2O _____ N_2O_4 _____

Complete the table below.

Formula of substance	Number of atoms of each element	Scientific name of substance	Element or Compound
$CaCl_2$	Calcium x 1 Chlorine x 2	Calcium chloride	Compound
CO_2			
	Sodium x 1 Chlorine x 1		
H_2			
	Hydrogen x 2 Oxygen x 1		
SO_2			
	Nitrogen x 2		
	Magnesium x 1 Oxygen x 1		
	Oxygen x 2		